SUMMANDS OF A REPRESENTABLE SHEAF ARE OPEN AND CLOSED SUBSCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short: the first step is a citation to the coproduct lemma, the second is the pointwise test that upgrades the isomorphism of sheaves to a coproduct decomposition of schemes.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Structure of a coproduct decomposition of a representable sheaf. Lemma lemma-coproduct-sheaves-open-and-closed, used at proof lines 1238-1240. Makes each F_i -> U representable by open and closed immersions.

Source reference. Original lemma 0F15: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18785–18785. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $S \in \mathrm{Ob}(Sch_{fppf})$. Let $U \in \mathrm{Ob}((Sch/S)_{fppf})$. Given a set I and sheaves F_i on $\mathrm{Ob}((Sch/S)_{fppf})$, if $U \cong \coprod_{i \in I} F_i$ as sheaves, then each F_i is representable by an open and closed subscheme U_i and $U \cong \coprod U_i$ as schemes.*

Proof. By Lemma 0F15 the map $F_i \to U$ is representable by open and closed immersions. Hence F_i is representable by an open and closed subscheme U_i of U. We have $U = \coprod U_i$ because we have $U \cong \coprod F_i$ as sheaves and we can test the equality on points. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.